Distrust memo: what in the Westline pack not to trust

Westline Carriers TPMS assessment. Each point says what is wrong or weak, and what I did about it in the analysis.

1. The two vehicle IDs are one truck.

KA25AB4471 and KA25B4471 overlap on 30 June and 1 July (28 slot-days). It looks like a typo in the July export, and the overlap hints at a time-base mismatch between the June and July files, although the data dictionary says all times are IST. *What I did:* relabelled both to one ID for every comparison. *Not done:* removing duplicate rows, because I only had daily tables, not raw rows.

2. Sensor labels may not follow the tyres.

The fleet manager says a pair was swapped left to right at the end of June but cannot say which truck. TPMS slots are provisioned to sensors, so a rotation can silently move a tyre's history to the wrong slot. *What I did:* made no claim that depends on one slot's history across late June. The register's "re-fitted" entries were not available to me, so this is unresolved.

3. The register's serials are tyre serials, not sensor serials.

The data dictionary defines WL-<fleet>-<position>-<nnnn> as the tyre serial. Earlier drafts called WL-T2-5RI-0242 a sensor. *What I did:* corrected the wording. The sensor battery jump (about 48% to 98%) on 12 July is the independent evidence for a sensor change.

4. The cold pressure standard cannot be compared to running readings.

TPMS values mix cold, warm and hot tyres, and the spec is a cold figure. A fixed-threshold alarm will fire whenever tyres cool. *What I did:* used idle Sundays (sensor temperature about 28 C) as the cold baseline, and did not use the standards as fault thresholds.

5. My daily tables are approximate.

They average chunk-level medians and shares, and take the minimum of chunk-level 10th percentiles, so they are not exact daily statistics. *What I did:* used them only to compare trucks and spot steps, and flagged every figure as approximate. Exact values need recomputing from raw rows.

6. The fleet manager is one informant, and partly wrong or unsure.

He attributes low pressure to Hubballi roads, but the pattern fits hot inflation; he cannot name the rotated truck; he describes the failure as having no warning, yet the data shows a decline from at least 10:00. *What I did:* tested his leak and no-warning claims against the data; the rotation claim could not be tested without the register.

7. Tread-depth sheets are too thin for wear conclusions.

Only the Goa truck was inspected twice; the Mangaluru and Vijayapura trucks never. Inspector codes (C, D, F1, R1) differ from TPMS codes, and they conflict: the dictionary says C is right and D is left, yet the manager says C is the cleaner's side, on the left, and in India the driver sits on the right. *What I did:* made no wear claims, and no tread-to-TPMS position matching, until this is confirmed.

8. Missing or unreliable inputs.

There are no load records, no hourly fill times, and no description of how TyreSense decides to alert. The manager distrusts the GPS ("jumps around"), and the kilometre log depends on phoned-in readings. *What I did:* left load, route and GPS out of every conclusion, and listed them as questions.

9. Empty slots are not failures.

Axle 6 and 7 slots (eight of 22) are empty on every truck by design. *What I did:* excluded them, so 14 slots per truck are analysed.